

# HamidReza Saadi Dadmarzi

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## Education

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| <b>Koç University</b> , M.Sc. in Computer Science                                                                  | 2023–Present |
| Research focus: Password-based authentication systems (MPC), privacy-preserving ML, and post-quantum cryptography. |              |
| <b>Amirkabir University of Technology(Polytechnic of Tehran)</b> , M.Sc. in Applied Mathematics                    | 2017–2020    |
| Research focus: Quantum Error-Correcting Codes and Information-Theoretic Cryptography.                             |              |
| <b>Amirkabir University of Technology(Polytechnic of Tehran)</b> , B.Sc. in Applied Mathematics                    | 2014–2017    |
| <b>Amirkabir University of Technology(Polytechnic of Tehran)</b> , Minor in Physics                                | 2016–2018    |

## Publications & Projects

- **UpSPA / RoUpSPA** ([ePrint](#), [code](#), [extension](#)): Lead author of *UpSPA* (submitted to ESORICS 2026) and *RoUpSPA* (under preparation for ACM TOPS). Designed threshold-based password authentication protocols and an open-source password-manager/browser-extension prototype. Proposed PBB-, quorum-, and blockchain-based synchronization architectures, together with Layer-1 and Layer-2 mechanisms for robust and scalable credential management.
- **FLDP / SpectraDP**: ongoing PPML work on frequency-domain differential privacy for federated learning; comparing spectral masking with random DP baselines under accuracy, privacy, communication, and reconstruction-error metrics.
- **CURE**: implementation/debugging support for homomorphic-encryption-based privacy-preserving split learning; helped investigate Go/Lattigo and parallel encrypted-experiment issues. Contribution was code-level support, not paper authorship.
- **QuasarSSO**: post-quantum password-protected SSO system under preparation.
- **AEGIS**: mechanizable framework for post-quantum IoT authenticated key exchange.

## Research & Professional Experience

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|--------------------------------------------------------------------------------------------------|--------------|
| <b>KU-Crypto Lab</b> , Koç University, Research Assistant                                        | 2023–Present |
| Working on cryptographic protocols and secure authentication systems under Prof. Alptekin Küpçü. |              |
| <b>BTL Coin</b> Software Developer                                                               | 2021–2023    |
| Developed automated crypto-asset management and market analysis tools using Golang and Docker.   |              |
| <b>Institute for Research in Fundamental Sciences (IPM)</b>                                      | 2020–2021    |
| Worked on quantum machine learning implementations using Qiskit and PyQuil.                      |              |

## Teaching & Academic Service

- **Teaching assistant for courses in**: Modern Cryptography, Linear Algebra, Computer Networks, Data Structures & Algorithms, Discrete Mathematics, Logic, Probability, PDE, General Physics II, and Calculus I–II at Koç University and Amirkabir University
- **External reviewer for**: IEEE TIFS (2024–present), ACM ASIACCS (2024, 2026), ESORICS (2025, 2026), ACNS (2026), Royal Society Open Science (2024), PeerJ (2026), JISA (2025), and Turkish Journal of Electrical Engineering & Computer Sciences (2025)

## Honors & Awards

- Outstanding Teaching Assistant Award, Koç University (2025).
- Exceptional Talent Center recognition for graduate admissions and academic performance.
- Ranked among the top 5% of undergraduate students.

## Technical Skills

**Programming**: Rust, Go, Python, MATLAB  
**Tools**: Docker, L<sup>A</sup>T<sub>E</sub>X, Qiskit, PyQuil, Lattigo  
**Areas**: Cryptography, SMPC, PQC.

## Conferences & Workshops

- Frontiers in Mathematical Science 5th,6th Conference - in IPM(Tehran) - July 2017,2018
- Workshop on Mathematics of Information-Theoretic Cryptography - in **IMS**, Singapore - September 2016

## References

- Prof. Alptekin Küpçü([link](#),[email](#))
- Dr. Erfan Salavati([link](#),[email](#))
- Prof. Masoud Pourmahdian([link](#),[email](#))